Financial Claim Verification and SEC Fact-Checking System

End-to-End Automated Pipeline

TECHNICAL OVERVIEW | ENGINEERING ARCHITECTURE

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Converting Unstructured Speech into Verifiable Financial Truth

OBJECTIVE

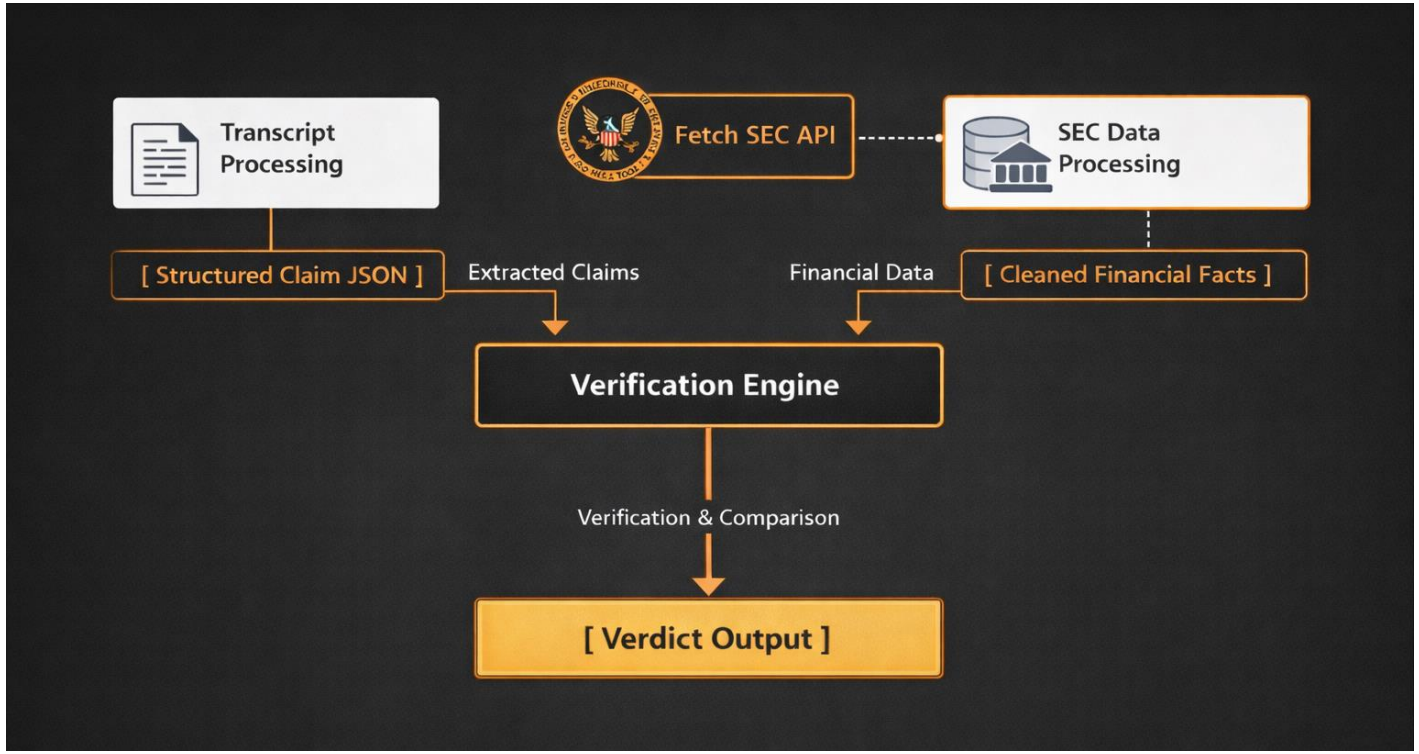
1. Ingest earnings call transcripts
2. Extract quantitative financial claims
3. Verify against official SEC regulatory filings

CORE CHALLENGE

- Moving beyond simple extraction to reliable, explainable verification
- Bridging the gap between executive rhetoric and regulatory ground truth

SPEECH TO TRUTH PIPELINE

System Architecture Overview



Key architectural decision:

- Transcript processing and SEC processing are independent
- They only interact at the verification layer
- This separation ensures modularity, testability, and scalability

Track A: Transforming Text into Structured Claims

PURPOSE
Convert raw, verbose, unstructured earnings call text into structured, machine-verifiable financial claims

Raw text

"Co. reported 1Q17 revenues of $78.4b, net income of $17.9b and diluted EPS of $3.36. Expects 2Q17 revenues to be $51.5-53.5b."


```
{
  "sec_metric": "us-gaap:Revenues",
  "kind": "absolute",
  "value": 78.4,
  "unit": "USD",
  "frame": null,
  "raw_sentence": "Co. reported 1Q17 revenues of $78.4b",
  "confidence": "medium"
},
```


What's working well

- Extracts explicit numeric claims reliably.
- Handles unit normalization (million vs billion).
- Distinguishes growth vs absolute value.
- Preserves context for explainability

Limitations

- Vague claims ("strong growth") are not verifiable.
- Non-GAAP phrases require deeper modeling.
- Some claims lack explicit timeframe.

Track B: Ingesting the Regulatory Source of Truth

SOURCE
SEC CompanyFacts API

FORMAT
Machine-Readable XBRL Data

STANDARD
GAAP Concepts

Rigid Backbone of Facts

The system fetches:

- CompanyFacts JSON
- Relevant US-GAAP metrics
- Period start/end dates
- Units and filing context

These records are cached locally

Filtering and Normalizing Noisy SEC Data

01

Form Filtering

Restricting ingestion to relevant 10-K and 10-Q forms

02

Period Filtering

Isolating specific fiscal years and quarters

03

Unit Normalization

Standardizing scale (millions vs. billions), USD, % values, share counts and format

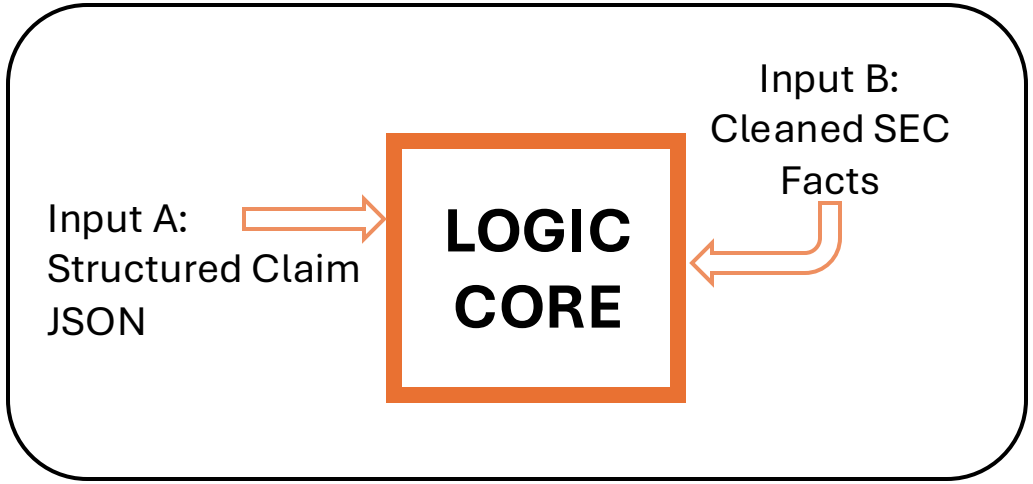
04

Frame Alignment

Syncing transcript quarters with SEC period end dates

Output is a clean, queryable financial fact layer – Used by Verification Engine

The Verification Engine: Where Logic Meets Data



Step 1: Metric Canonicalization

Transcript phrases are mapped to canonical SEC tags

- Net income → NetIncomeLoss
- Cash on hand → CashAndCashEquivalentsAtCarryingValue

Step 2: Select the Best SEC Data Point

- Searches all candidate tags
- Filters by correct unit type and matches exact period end when possible
- Prefers consolidated-level facts over segment-level or mismatched dimensions

This reduces false inaccuracies caused by wrong tag selection

Step 3: Verifying Absolute Values and Growth

A) Absolute Value Claims

Compare Claim Value vs. SEC Value

Logic: Apply relative tolerance checks to account for rounding

B) Growth Claims (YoY/QoQ)

1. Retrieve Current Period Value
2. Retrieve Prior Period Value
3. Compute Delta: (Current – Prior) / Prior
4. Compare Computed Growth VS Claimed %

C) Derived Metrics

Computing composite metrics to reduce 'Unverifiable' verdicts

Cash + Marketable Securities

The Output: Verdict, Delta, and Explanation

1. Verified
2. Inaccurate
3. Unverifiable

```
==================================
  VERIFICATION RESULTS: AAPL
==================================
  Total claims:  181
  VERIFIED:      38
  INACCURATE:    110
  UNVERIFIABLE:  33
==================================
```

VERDICT: VERIFIED


```
{
    "quarter": "Q1",
    "verdict": "VERIFIED",
    "sec_metric": "us-gaap:Revenues",
    "kind": "absolute",
    "claim_value": 78.4,
    "claim_unit": "USD",
    "sec_value": 78.351,
    "sec_unit": "USD",
    "delta": 0.049000000000006594,
    "delta_pct": 0.06253908692933924,
    "explanation": "SEC value: 78.35 USD.",
    "raw_sentence": "$78.4b"
  },
```

System Evaluation: Validated Success

End-to-End Automation → Reliable flow from Transcript to Verdict

Unit Handling → Million/Billion mismatches resolved correctly

Transparent Explanation → Every verdict includes reasoning and source data

Improved Tag Matching → Advanced selection logic reduces false negatives

Growth Calculation working → Pulls current and prior SEC values, calculates YoY or QoQ growth, verifies it against the claimed percentage, and allows rounding tolerance

System Evaluation: Current Limitations

1 Growth Fragility

- If prior-period SEC value is unavailable, growth cannot be computed
- Missing frames or mismatched fiscal periods cause UNVERIFIABLE/INACCURATE verdicts

2 Metric Mapping Gaps

- Some transcript phrases do not map cleanly to GAAP tags
- Ambiguous mapping leads to UNVERIFIABLE/INACCURATE verdicts

3 Non-GAAP

- “Adjusted EPS” or “Revenue excluding FX” require reconciliation tables
- SEC CompanyFacts only contains GAAP values
- System does not model non-GAAP adjustments

4 Fiscal Quarter Alignment

- Companies have different fiscal year ends
- Was working on fix but not yet solved completely

5 Vague Phrasing

Subjective claims like “Strong growth” remain outside scope

Hybrid Approach for Claim Extraction

Earnings Transcript

- Unstructured
- Conversational
- Inconsistent

LLM = Language Understanding

- Detects financial claims
- Understands varied phrasing
- Classifies: Absolute | Growth | Margin
- Extracts timeframe

Rules Based Normalization

- Parse numbers deterministically
- Normalize units
- Validate % and timeframe
- Output clean structured JSON

```
[2017-Aug-01-AAPL.txt] Extracting (mapped to Q3)...
      → 67 claims extracted

[2017-Jan-31-AAPL.txt] Extracting (mapped to Q1)...
      → 47 claims extracted

[2017-May-02-AAPL.txt] Extracting (mapped to Q2)...
      → 38 claims extracted

[2017-Nov-02-AAPL.txt] Extracting (mapped to Q4)...
      → 29 claims extracted
```

LLM interprets meaning

Rules validate numbers

Result: Flexible understanding + deterministic accuracy

Final Conclusions

1. Built an End-to-End Financial Claim Verification System
2. Hybrid Architecture Enables Both Flexibility and Accuracy
3. Explainable, Transparent Outputs
    - Shows SEC tag used
    - Displays official value
    - Calculates delta
    - Provides clear reasoning for each verdict
4. Handles Multiple Claim Types
    - Absolute values, Growth (YoY / QoQ), Margins, Derived financial metrics
5. Modular and Extensible Design
    - Independent transcript and SEC pipelines
    - Verification logic can scale
    - Future enhancements possible

GitHub Link

Architecture is modular, extensible, and transparent

Future Scope

Currently, verification is deterministic and rule-based. Future improvements could incorporate LLM support in specific areas.

Context-Aware Claim Classification

LLMs could help:

- Distinguish GAAP vs Non-GAAP claims
- Detect when executives are referencing segment-level vs consolidated numbers
- Identify guidance vs reported results

This would reduce false UNVERIFIABLE classifications.

Non-GAAP Reconciliation Modeling

LLMs could:

- Parse reconciliation tables in filings
- Identify adjustment components
- Help reconstruct adjusted metrics

This would allow verification of “Adjusted EPS” or “Revenue excluding FX”

Improved Metric Mapping

LLMs could assist in:

- Mapping ambiguous transcript phrases to correct GAAP tags
- Understanding nuanced phrasing (e.g., “core operating earnings”)
- Suggesting likely tag candidates when rule-based mapping fails

This strengthens the transcript → GAAP mapping layer

Confidence Scoring

LLMs could assist in:

- Assessing ambiguity in metric mapping
- Evaluating contextual clarity
- Supporting probabilistic confidence scoring per verdict

Even with LLM enhancements, numeric truth comparison will remain deterministic and rule-based

THANK YOU